



Figure 25-9 A dog with contracture of the left carpus and digits after immobilization of the forelimb as a result of a fracture. Note the upright stance of the foot in comparison with the normal contralateral side.

the limb and perform the stretch with the joints in the proper alignment to reduce any abnormal stresses on the joints. After the stretching period, the tissues are allowed to return to a neutral position, and then the stretch is reapplied for up to 20 repetitions in a session.

Prolonged Mechanical Stretching

Prolonged mechanical stretching is similar to static stretching in that a low-intensity stretch is applied. It differs, however, in that the stretch is prolonged. The stretch, applied for a minimum of 20 minutes and up to several hours, is effective in increasing ROM in many patients.²³ In people, a low-intensity prolonged stretch applied 1 hour per day was more effective than static stretching over a 4-week period in patients with chronic knee flexion contractures.²⁴

In animals, an effective manner of achieving prolonged mechanical stretching is the use of splints or other coaptation devices to provide prolonged stretch to tissues. For example, dogs with a fracture of the radius and ulna treated with a cast often have contracture of the carpus following fracture healing (Figure 25-9). Pain and reduced limb use are frequent sequelae. A dorsal fiberglass splint, which has been contoured to the normal contralateral side, may be placed on the affected limb over several layers of cotton cast padding that have been placed over the limb. Gauze bandaging material may then be used to help pull the foot up to the contoured splint with mild tension, applying controlled prolonged gentle stretch to the palmar carpal

tissues. The wrap may be changed several times per day, and static stretching and ROM exercises performed. When the wrap is replaced, additional stretch may be applied by increasing the tension on the gauze bandaging material. At no time should the patient feel undue discomfort. After several days of this form of treatment, the carpus generally returns to a normal ROM and function and pain are improved.

Other devices to provide prolonged mechanical stretch in veterinary medicine include the use of a dynamic flexion apparatus to apply a flexion stretch to the stifle in patients with fracture disease and quadriceps contracture.^{25,26} In some instances, it may be necessary to surgically release adhesions and scar tissue. A dynamic flexion apparatus may then be applied to help prevent recurrence of the contracture. The apparatus consists of transcutaneous pins placed in the pelvis or proximal femur, and in the tibia, with elastic bands placed between the pins to provide prolonged stretch with the stifle in flexion. In addition, the patient may also actively extend the stifle against the resistance of the elastic band, but when the quadriceps muscle relaxes, the elastic band applies stretch to the stifle in flexion.

It is believed that when muscles are maintained in a lengthened position for several weeks, sarcomeres are added.^{20,27} This increase in length persists if the new length is actively used. Other connective tissues may also lengthen as a result of plastic deformation of the tissue.

Ballistic Stretching

Ballistic stretching (sometimes referred to as *dynamic stretching*) differs from static stretching in that a series of movements are used to stretch the muscles and connective tissues.² This is a high-intensity, short-duration bouncing stretch.¹ The stretch that occurs often results in contraction of the antagonistic muscles to the tissues being stretched. Although effective stretching may occur with this form of stretching, there is a risk of tissue injury, and this form may not be appropriate for patients early in the postoperative period. In fact, the muscle tension that occurs with ballistic stretching is approximately twice that obtained with low-intensity prolonged stretching.²⁸ In general, a period of light warm-up activity and static stretching exercises should precede ballistic stretching. In humans, ballistic stretching is sometimes used by individuals who need to perform quick motions such as in martial arts or hurdling. There may be similar applications in canine athletes, but this has not been investigated.

Proprioceptive Neuromuscular Facilitation Stretching

Proprioceptive neuromuscular facilitation stretching is more complex and takes advantage of neuromuscular stretch activation. The stretches use a contract-relax

sequence, an agonist muscle contraction, or a contract-relax-agonist muscle contraction sequence.² Unfortunately, these stretches require the patient to actively and consciously contract a muscle group. Because we cannot verbally communicate to an animal when the muscle should contract, the therapist must use the other forms of stretching. There is a possibility that neuromuscular electrical stimulation may be used to provide active muscle contraction of a particular agonist muscle group, but clinical experience in this technique is limited.

Precautions and Contraindications to Stretching

Joints should not be forced into an uncomfortable position beyond the available ROM. Patients should not be painful during or after stretching. Caution should be exercised if a limb has been immobilized for a length of time. The goal should be to stretch and realign connective tissue, not rip and tear tissues. Stretching should not be performed if a ligament or tendon has been injured until the fibrous tissue can withstand some stress. Similarly, caution should be exercised during stretching in regions of relatively new fractures after repair has been performed.

Stretching Studies

Although there is a relative paucity of data and scientific knowledge regarding stretching to improve flexibility, especially in animals, it is a widespread practice in both human and animal rehabilitation. A study in Labrador retrievers with elbow and stifle osteoarthritis investigated the effects of a home passive stretching program carried out by owners on joint ROM. Owners were carefully instructed on how to perform the stretches and received supervised practice. ROM measurements were obtained at the start of the study and after a 21-day home program. The home program consisted of owners performing 10 stretches of 10-second duration each two times daily. After 21 days, there was a significant increase in ROM, with an average increase of 14.6 degrees.²⁹

Studies of stretch behavior of normal human muscle have yielded interesting results. One study evaluated the passive properties of the hamstring muscle group during stretch while measuring passive hamstring resistance and electromyographic activity.³⁰ Resistance to stretch was defined as passive torque of the hamstring muscle group during passive knee extension. The knee was passively extended to a predetermined final position where it remained stationary for 90 seconds (static stretch). Alternatively, the knee was extended to the point of discomfort (stretch tolerance). Passive energy and stiffness were calculated. Torque decline in the static phase was considered to result from viscoelastic stress relaxation of the tissues. A single static stretch resulted in a 30% viscoelastic stress

relaxation. With repeated stretches, muscle stiffness temporarily decreased, but returned to baseline values within 1 hour. Stretching over a 3-week period increased joint ROM because of a change in stretch tolerance rather than in the passive properties of the muscle. Strength training resulted in increased muscle stiffness, which was unaffected by daily stretching. Inflexible and older subjects have increased muscle stiffness, and a lower stretch tolerance compared with subjects with normal flexibility and younger subjects. The concept of an increase in stretch tolerance was confirmed in another study of people in which individuals underwent a stretching protocol of the hamstring muscles for 6 weeks.³¹ Although there was no increase in the extensibility of the hamstring muscles, there was an improvement in stretch tolerance, which resulted in an increase in joint ROM.

Another study evaluated the effects of one 10-minute stretch on muscle stiffness in people with short hamstrings.³² The subjects were unable to touch the floor when bending forward. One group performed static stretching exercises for 10 minutes interspersed with relaxing, and an untreated group served as controls. One 10-minute stretch resulted in a significant increase in passive muscle moment, ROM, and elongation of the hamstrings. There was no significant change in passive muscle stiffness, however, with respect to the prestretch stiffness curve. The authors also concluded that one session of static stretching does not influence muscle stiffness, and the increased ROM and extensibility of the hamstrings resulted from an increase in stretch tolerance.

Acute changes in the flexibility of hamstring muscles were evaluated in older adult humans.³³ A single 32-second stretch was performed by men and women with a mean age of 65 years. Although the changes were moderate, the increase in flexibility was significantly greater in the stretch group as compared with the controls (4 degrees vs. 1 degree).

Although most muscle stretching studies have focused on defining the biomechanical properties of isolated elements of the muscle-tendon unit or on comparing different stretching techniques, one group of investigators evaluated stretching properties in an entire muscle-tendon unit.³⁴ Rabbit long digital extensor and cranial tibialis muscle-tendon units were evaluated using common stretching techniques. The effect of varying stretch rates was also evaluated. Similar to other studies, the muscle-tendon units responded viscoelastically to tensile loads. Both cyclic and static stretching resulted in sustained muscle-tendon unit elongations, suggesting that greater flexibility can result if these techniques are used clinically. With repetitive stretching, there was little additional alteration of the muscle-tendon after four stretches, implying that a minimum number of stretches leads to most of the elongation in repetitive stretching. Also, greater peak tensions and greater energy absorptions occurred at faster stretch rates,

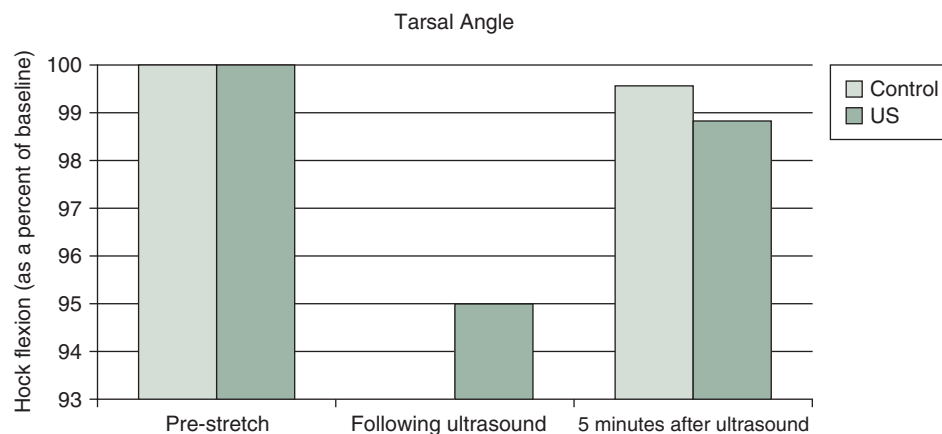


Figure 25-10 A mild stretch applied to the common calcaneal tendon immediately after ultrasound treatment (10 minutes of 3-MHz continuous ultrasound heating at 1.5 W/cm²) resulted in a 5-degree increase in hock flexion, but this increased flexion was lost within 5 minutes after the ultrasound was discontinued. Repeated stretching of control tendons without ultrasound resulted in no increase in hock flexion.

suggesting that the risk of injury in a stretching regimen may be related to the stretch rate. Indeed, one group of investigators found that during slow (or static) muscle stretching, factors inhibiting motor neurons contributed to the lengthening of the muscle.³⁵

Passive stretching may also help to prevent muscle atrophy by allowing constant change in the length of the muscle fibers. Intermittent muscle stretching helped to prevent atrophy and loss of sarcomeres in muscles of mice that were immobilized in a shortened position.³⁶ In this model, a static stretch for as little as 30 minutes per day helped to preserve sarcomeres. In addition, there was less development of intramuscular connective tissue with some stretching. In a study of rat tail tendons, a low-force stretch over a prolonged period resulted in effective elongation of the tendon.³⁷ Also, the proportion of prolonged tendon lengthening was greater with low-force, long-duration stretching.²³ Heating tissue before stretching has been advocated to help improve the effectiveness of stretching. Elevating the tissue temperature before stretching the rat tail tendons resulted in significantly less tissue damage; and lower loads applied with heating for a prolonged period resulted in significantly greater elongation. One researcher concluded that approximately three times as much force is needed to elongate tendons at temperatures of 20° to 30° C than at 43° C.³⁸

Therapeutic ultrasound is frequently mentioned as an adjunctive treatment to improve the effectiveness of stretching through its ability to warm tissues. However, the specific effects of heating on tissues and the manner in which stretch is affected is unclear. In a study of ultrasound treatment of the common calcaneal tendon of dogs, it was found that pulsed-mode ultrasound was not effective in increasing tendon temperature, but continuous-mode ultrasound was.³⁹ The amount of heating with 3-MHz continuous-mode ultrasound was related to the power, with 1.5 W/cm² increasing tendon temperature more than did

1 W/cm². Interestingly, the greatest temperature increase occurred within 3 minutes, with little increase thereafter. Tendon cooling occurred very rapidly, with return to baseline temperature within 1 minute after stopping ultrasound. A 4-kg stretch applied to the tendon immediately after ultrasound treatment (10 minutes of 3-MHz continuous ultrasound heating at 1.5 W/cm²) resulted in a 5-degree increase in hock flexion, whereas two bouts of stretching of the contralateral nontreated calcaneal tendon within 5 minutes did not result in increased hock flexion. However, the increased hock flexion was lost within 5 minutes after the ultrasound was discontinued (Figure 25-10). Clearly, methods of maintaining the increased stretch are needed to improve tissue extensibility.

One study performed on humans also evaluated the effects of (1) static stretch and ultrasound, (2) static stretch alone, or (3) no treatment on triceps surae muscle extensibility.⁴⁰ Static stretch and ultrasound consisted of 7 minutes of continuous ultrasound at 1.5 W/cm² to the muscle with stretch applied during the seventh minute. Static stretch alone consisted of a 51-lb load applied for 1 minute. Static stretch and ultrasound treatment increased elbow flexion an average of 1.2 degrees, or 20%, more than static stretch alone, whereas static stretch alone increased flexion an average of 1.3 degrees, or 27%, more than no treatment. Both increases were statistically significant.

A similar study on people confirmed the short-term benefit of heating with ultrasound before stretching.⁴¹ One group received ultrasound (1.5 W/cm²) to the musculotendinous junction of the triceps surae and then stretching, and another group received stretching alone. Ultrasound and stretch increased mean ROM more than did stretch alone, but there were no long-term residual effects.

Another study evaluated whether heating with continuous-mode ultrasound would augment the effect of stretching of the medial collateral ligament of the knee in normal humans.⁴² Treatment consisted of 3-MHz, 1.25 W/cm²

continuous ultrasound or sham treatment over the medial collateral ligament. A valgus stretch of the knee (10 ft-lb for 2.5 min) was performed after ultrasound treatment. Stretching alone resulted in a 1-degree valgus displacement of the knee, whereas stretching with ultrasound resulted in a displacement of 1.24 degrees. However, these differences were not statistically significant.

The effect of stretching on athletic performance such as agility competitions, field trials, Schutzhund, etc. is controversial. Traditionally, static stretching precedes athletic performance. However, the effectiveness of stretching on athletic performance has come into question in recent years based on research. Several studies have shown a decrease of muscle force output around a single joint associated with static stretching.⁴³⁻⁴⁵ Proposed mechanisms for this decrease in muscle force include a decrease in muscle stiffness,⁴⁵ muscle inactivation during muscle stimulation,⁴³ and decreased muscle activity after maximum voluntary contraction. In addition, because stretching can reduce muscle stiffness, it may have a negative effect on the ability to generate a forceful contraction.⁴⁵ Stretching also appears to decrease motor neuron excitation.⁴⁶ One study evaluated the effect of stretching on agonist and antagonist muscle activity and muscle force output with joint isometric contractions.⁴⁷ The authors concluded that stretching appears to decrease muscle force output by up to 20% in a single joint isometric contraction and the rate of force development in a multiple joint isometric contraction in people. Stretching may cause a decrease in muscle force generation up to 16 minutes after stretching and may not be recommended as a general warm-up procedure, especially prior to strength or power activities. Current recommendations for stretching during warm-up activities should be carefully considered. It is not clear how applicable these results would be when combined with warm up, practice drills, and low-intensity dynamic exercises.

Another traditional belief is that stretching prior to activity may result in fewer injuries associated with performance. This topic has been recently reviewed in people.⁴⁸ The purposes of stretching before athletic performance include ensuring that there is sufficient ROM in the joints to perform the activity, and to reduce muscle stiffness, theoretically reducing the risk of injury. The effect of stretching on injury prevention has shown mixed results. In general, it is believed that stretching along with warm-up does not affect the incidence of overuse injuries. However, stretching may reduce the incidence of muscle strains. In general, it may be beneficial to perform static stretching when a large ROM of various joints is required for activity. A combination of stretching and warm up is likely needed to reduce the risk of injury in these situations. It is most beneficial to target the stretching to those muscle groups known to be at risk for a particular activity. At least four to five 60-second stretches to the point of initial discomfort are recommended to reduce resistance to stretch, followed by some

sport-specific warm-up activity. However, stretching prior to an activity that has more overuse injuries rather than muscle strains is unlikely to be effective in reducing injury.

Clinical Application of Stretching

In preparation for stretching, it is beneficial to perform some low-intensity active exercise if possible, such as walking and light jogging. In addition, application of superficial heat or therapeutic ultrasound before stretching may improve tissue extensibility. There may be less damage to tissues, and less force may be required to attain increased tissue length with stretching if the tissues are warmed. The combination of muscles, tendons, skin, and joint capsule should be considered and their relative contributions to restricted motion assessed. For example, the target tissue for lengthening a muscle that is contracted is the muscle belly, not the tendon. The patient should be in a comfortable position, which is generally in lateral recumbency, and should be on a padded surface. It is important that the patient be as relaxed as possible, and, in some cases, mild sedation or tranquilization may be beneficial.

Static Stretching

One hand should stabilize the bone proximal to the joint, and the other should stabilize the bone distal to the joint. In general, the distal bone should be moved relative to the proximal bone. The affected limb and joint should be slowly placed through one end of a ROM (usually beginning in flexion) until a restriction to motion is felt. Very gentle traction may be applied to the joint while slowly stretching to the point of initial restriction. The patient may indicate that it feels mild discomfort, such as by turning the head or mildly tensing the muscles in the affected limb. Under no circumstances should more severe pain be inflicted, which might be indicated by vocalizing, trying to move away, or attempting to bite. The stretch should be prolonged, ideally for at least 15 seconds. During the stretch, a conscious effort should be made to try to increase the joint excursion without increasing the level of discomfort. There should be no bouncing motions during the stretch with static stretching. Following the stretch at one end of the motion range, the pressure is slowly released and the opposite end of the ROM should be stretched. In some situations, dogs that resist manual stretching may tolerate other stretching techniques, such as placing the dog in ventral recumbency with the limbs flexed under the body and slowly pressing down on the body to achieve greater elbow flexion (for example, [Figure 25-11](#)).

In some situations, muscles that cross two joints, such as the rectus femoris and biceps brachii muscles, may be restricted. In these situations, it is beneficial to put an individual joint through a stretching session, and then place the other joint through a stretching session. As ROM and flexibility improve, both joints may be simultaneously



Figure 25-11 A mild stretch is applied to increase elbow flexion using the dog's body weight. The antebrachium is comfortably supported on a table with padding.

placed through stretching exercise. For example, the hip and stifle joints may be stretched in extension. This is more challenging, and there will usually be less joint excursion with simultaneous extension of both joints than if each joint is individually stretched in extension. However, these motions more completely reflect the use of these joints and muscles during normal gait, and it is valuable to perform these maneuvers.

The therapist should be patient and not try to achieve full ROM in one or two sessions. The ideal daily frequency of stretching is unknown for dogs. In general, two to four sessions per day may initially be required, with the frequency decreased as normal ROM and tissue extensibility improve. The process must be applied consistently and regularly to obtain good results, and it may take 2 to 3 weeks to see noticeable improvement.

Prolonged Mechanical Stretching

The most problematic joints that are affected with immobilization of a limb in a cast or splint are the digits, carpus, and tarsus, which may develop flexural contracture. This may occur in either the front or the rear limb joints as a result of fractures of the radius and ulna or tibia and fibula, or in the digits with surgical fusion (arthrodesis) of the carpus or tarsus. After 4 weeks in a coaptation device, the joints are often contracted in a flexed position and are painful when attempts are made to place them in a functional walking position.

Normal static stretching of the affected joints should be performed. It is usually only practical to stretch all of the digits at the same time. Between stretching sessions, a dorsal fiberglass splint, molded to the normal contralateral side, is placed on the affected limb to help maintain a prolonged mechanical stretch. The splint may be manufactured using fiberglass casting material folded back and forth on itself, using the normal limb as a template. Several layers of cotton cast padding are placed over the affected

limb. The dorsal splint is applied, and gauze bandaging material is rolled on to the limb under mild tension to pull the foot up to the splint, applying controlled prolonged gentle stretch to the restricting tissues. The wrap is changed one to three times per day, and static stretching and ROM exercises should be performed while the wrap is removed. When the wrap is replaced, additional stretch may be applied by increasing the tension on the gauze bandaging material. In general 5 to 10 days is usually adequate to regain normal ROM of the digits. Similar principles may be applied to more proximal joints, such as the elbow and stifle. In some situations, it may be difficult to use a fiberglass splint, and placement of transcutaneous pins in the bones proximal and distal to the affected joint may be necessary to provide good fixation. Elastic bands may be placed to achieve flexion or extension as needed.

Ballistic Stretching

To perform ballistic stretching, the patient should be comfortable and the affected limb stabilized. The joints are positioned until a gentle stretch of the target tissues occurs, and then the therapist applies gentle “bouncing” of the tissues. The therapist should be careful to use only gentle movements to avoid damage to the tissues and pain. Because of the risk of tissue injury, this technique should not be performed in traumatized tissues, after surgery, or on tissues that are inflamed or edematous.

Ballistic stretching may be beneficial to normal dogs that undergo ballistic-type activities, such as agility dogs that jump and turn sharply, or dogs that play Frisbee. A technique of ballistic stretching in sporting dogs may be performed using a therapy ball or roll, trampoline, or inflatable mattress. The trampoline and mattress provide greater stability so that a single person can perform the stretches, whereas two people are generally needed when a therapy ball or roll is used. With the animal in a standing position, the animal is gently bounced up and down. Depending on the situation, the dog may also be placed in a sternally recumbent position on the device. Gentle bouncing of the dog will allow ballistic stretching of the joints in a flexed position. Ballistic stretching may also be performed with a joint in extension using a therapy ball or roll. For example, a dog may have its forelimbs on the ground with the rear limbs pulled back in extension over a therapy roll (Figure 25-12). Rapid, gentle rolling of the therapy roll back and forth or gentle bouncing of the rear limbs on the roll results in ballistic stretching of the hip and stifle joints. Similar principles may also be used for ballistic stretching of the forelimb joints in extension.

Summary

ROM and stretching exercises are basic parts of a complete physical rehabilitation program. The goals of ROM and stretching should be to lengthen muscles, improve joint

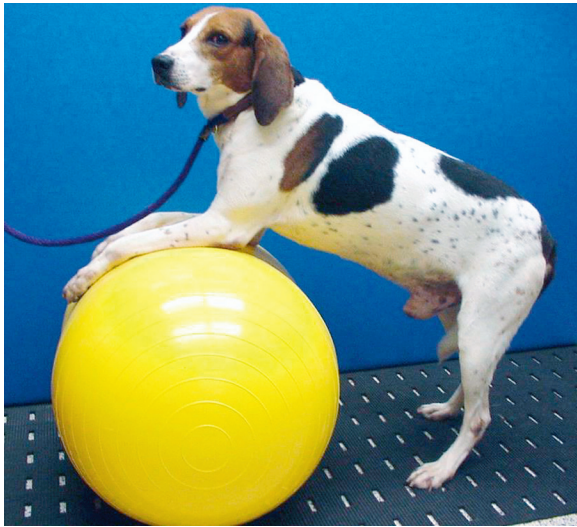


Figure 25-12 A physioball is used to increase hip extension by gradually rolling the ball forward to the point of a stretch on the cranial aspect of the hip joint and the periarticular tissues.

ROM, increase tissue mobility, and improve functional use of the affected joints and limb.

The therapist must be patient and be satisfied with small gains and slow improvement. To properly monitor patients, frequent measurement of ROM should be obtained using goniometry, and the results should be recorded to chart the patient's progress. Impatience and overzealous treatment may ultimately slow improvement and may actually reduce joint motion and worsen contracture of soft tissues. This is because tissue damage and pain may delay the active use of a limb through the available ROM, resulting in increased fibrosis of tissues and decreased flexibility.

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26 Joint Mobilization

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Manual therapy techniques are skilled hand movements intended to improve tissue extensibility, increase range of motion (ROM), induce relaxation, mobilize or manipulate soft tissue and joints, modulate pain, and reduce soft tissue swelling, inflammation, or restriction.¹ The primary techniques included in manual therapy are mobilization and manipulation of joints and associated soft tissues. Mobilizations are passive movements that are either oscillatory or sustained stretch performed in such a manner that the patient can prevent the motion if so desired. These motions are performed anywhere within the available ROM. The intent of this chapter is to provide an overview of the principles of manual therapy, followed by selected treatment techniques for the hip, stifle, elbow, shoulder, carpus, and the thoracic and lumbar spine. The techniques of GD Maitland,^{2,3} an Australian physical therapist who developed a clinically based approach in the 1960s and 1970s, are emphasized in this chapter. Maitland described four grades of mobilization (I-IV) (Figure 26-1) and manipulation. Manipulation, a grade V mobilization, is a high velocity, low amplitude passive movement that cannot be prevented by the patient and is typically performed near the end of the available ROM.

There have been numerous randomized controlled trials (RCTs) in humans that have demonstrated the efficacy of manual therapy for treating patients with a variety of disorders in the spine and peripheral joints. Many of these studies have compared “traditional treatments,” such as exercise, pharmaceutical interventions, rest, and placebo, with manual therapy. Other RCTs have compared these traditional forms of therapy to groups receiving both the traditional therapies plus manual therapy. The majority of these studies have found that manual therapy is as effective if not superior to traditional therapies.⁴⁻¹⁵ Common outcome measures assessed have been pain, functional scales, disability levels, ROM, number of treatments needed, length of time in treatment, and cost effectiveness of treatment. Despite the growing body of evidence for manual therapy, it is viewed as a complementary therapy in human medicine by many, even though its effectiveness seems well substantiated and the risks are low. The most likely reason for its slow acceptance is that the skill level required to

properly apply these techniques is higher than with traditional therapies such as exercise or modalities. Rationales for the reasons manual therapies may work have been investigated and include reducing muscle inhibition,¹⁶ decreasing pain,^{17,18} improving intervertebral disk hydration,¹⁹ correcting joint displacement, adjusting joint subluxations, restoring bony alignment, reducing nuclear protrusion,^{20,21} and placebo effect.⁸ More recent theories include evidence supporting the need for adequate stresses and normal movement as being critical to maintain the integrity of collagenous tissues, muscles, and bones.²² Evidence of the effectiveness of manual therapy must be established in small animals. Although many anatomic similarities exist between humans and small animals, we cannot assume the techniques will yield the same results.

Indications for Manual Therapy

Manual therapy is indicated for pain and loss of motion that occur secondary to neuromusculoskeletal dysfunction. The patient may have pain with motion or rest; pain in the midst of available ROM or at the end of available ROM; or pain or stiffness resulting from postural changes. Specific examples include dogs with limited motion secondary to canine hip dysplasia, elbow dysplasia, intervertebral disk disease, and osteoarthritis. Manual therapy differs from stretching in that when a stretch is applied, a low load is placed on the tissues for a specified amount of time (usually 10 to 30 seconds) to help elongate them. In manual therapy the force is applied in an oscillatory manner rather than in a sustained manner.

Contraindications and Precautions for Manual Therapy

The list of contraindications presented by Dutton²³ includes spinal instability, bacterial infection, malignancy, systemic localized infections, sutures over the area, recent fracture, cellulitis, febrile state, hematoma, acute circulatory condition, an open wound at the treatment site, osteomyelitis, advanced diabetes, hypersensitivity of the skin, inappropriate end feel (spasm, empty, bony), constant severe pain,

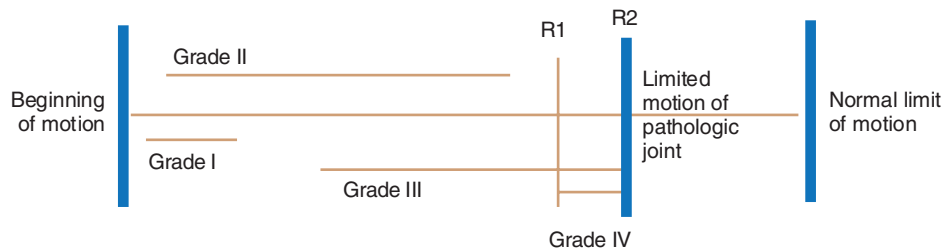


Figure 26-1 Graphic description of grades of joint mobilization.

extensive radiation of pain, pain unrelieved by rest, any undiagnosed lesion, and severe irritability (pain that is easily provoked, and that does not go away within a few hours). Additional contraindications specific to dogs include contractures such as fibrotic myopathy and quadriceps contracture, or in other circumstances in which manual therapy is unlikely to effect any change and may be painful. An overly aggressive or fearful dog that may bite the therapist or unrelaxed dogs are also contraindications. Total elbow and hip replacements should be considered a contraindication for grades III-V accessory mobilizations until the appropriateness of joint mobilization techniques are further evaluated for these procedures.

Precautions (proceed with caution) identified by Dutton²³ include joint effusion or inflammation, rheumatoid arthritis, presence of neurologic signs, osteoporosis, hypermobility, pregnancy (if the technique is to be applied to the spine), and steroid or anticoagulant therapy.

Basic Principles of Manual Therapy

There are a number of approaches to manual therapy, including those developed by physicians²⁴⁻²⁶ and by physical therapists.^{2,3,27-29} Regardless of the approach, there are several principles that should be considered as the patient is examined and treated. Maitland stresses the importance of communicating with the patient to fully “understand what the patient is enduring.”² Of course, in the case of animals, the examiner must communicate with two constituents—the dog and the owner. Body postures, pain patterns, and respiratory patterns should be closely monitored in the patient.

Physiologic and Accessory Motion

Physiologic motion is the normal active motion that is available at any synovial joint. Another way to describe physiologic motion is the motion that occurs in the cardinal planes. Examples include flexion, extension, abduction, and internal rotation. Accessory motions are movements that cannot be performed actively, but can be performed passively. Examples are distraction, glides, spins, and rotations of a joint. Accessory motions must be present for full physiologic motion to be present.

Concave and Convex Relationship of Joints

All synovial joints have a convex-concave relationship. When the examiner is passively moving a joint, caution should be made to move the joint in a manner similar to how it moves when the joint is being actively moved by the dog. Osteokinematics is defined by how the bone is being moved through space (i.e., flexion, abduction). Arthrokinematics is defined by how the joint surfaces are moving as the bone is being moved (i.e., rolling, sliding, and spinning). When the joint surface is convex with respect to the other concave side of the joint, the articular surface moves in the opposite direction of the shaft of the bone. When the shoulder joint is being flexed (as in the swing phase of gait) by moving the humerus on the scapula, the convex surface of the proximal humerus is sliding and spinning on the concave glenoid of the scapula. When the joint surface is concave, the articular surface moves in the same direction of the shaft of the bone. When the distal radius is being moved on the stationary carpal bones (as in the stance portion of gait), the concave surface of the radius is rolling and sliding on the convex proximal row of carpal bones. Manual therapists strive to move joint surfaces physiologically to avoid injuries such as joint subluxations and sprains.

Relationship of Pain and Range of Motion

Maitland states, “During examination and assessment, pain should never be considered without relation to range nor range without relation to pain.”² Depending on the nature of the injury or disease, joints are affected by the presence of pain and stiffness. If pain is the primary problem, it may limit motion, as in hip dysplasia. When the pain diminishes, the ROM may improve. If the primary problem is joint dysfunction or stiffness, pain will be most evident at or near the end of available ROM. As the available ROM increases, pain will be less of a factor. In those cases in which both pain and loss of motion occur simultaneously, the examiner must decide which is the primary problem. These relationships are the basis of deciding which grade of passive movement and other therapies should be used to treat the problem.

Table 26-1 Joint End Feel Characteristics

Name	Description	Example
Bony or hard	Bone approximates bone, resulting in an abrupt, hard stop.	Loss of flexion with end-stage elbow osteoarthritis; loss of stifle motion as a result of quadriceps contracture following fracture of the distal femoral physis
Soft tissue approximation	Motion is stopped by compression of soft tissues. Abnormal if occurs too early in the range as the result of edema.	Normal stifle flexion
Capsular or firm	A firm but slightly yielding stop occurs as a result of tension in the joint capsule or ligaments. Abnormal if occurs too early in the ROM.	Normal carpal extension
Springy block	A rebound is felt at the limit of motion—motion stops and then rebounds. Abnormal; may indicate joint effusion or a joint mouse.	Entrapped torn medial meniscus.
Empty	No end point is felt because the patient stops the motion because of pain; no resistance felt. Abnormal; indicates presence of sharp pain.	Sciatic nerve entrapment as a result of a migrating intramedullary pin; fracture of the lateral humeral condyle

ROM, Range of motion.

If pain occurs before the examiner feels resistance, the primary problem is pain; if the pain occurs when resistance is felt, or after, the primary problem is limitation of motion with joint involvement. In addition, if active and passive motions are limited or painful in the same direction, the lesion is in the noncontractile tissues. For example, if the cranial joint capsule of the stifle is tight, flexion of the stifle is limited with active and passive flexion of the joint. Conversely, if active and passive motions are limited or painful in opposite directions, the lesion is in the contractile tissues—muscle and associated tendon. For example, if the biceps tendon is affected, active flexion of the elbow and passive extension of the elbow are painful with a possible reduction in ROM. These presentations affect which grade and type of mobilization should be used to treat the problem.

The dog exhibits a *capsular pattern* if the entire capsule of the joint is inflamed or involved. Cyriax²⁴ described a capsular pattern as a joint-specific pattern of loss of motion with arthritis. These have not been described for dogs, but limitations in movement accompany certain disorders. An example is that hip extension and hip abduction are typically the most limited motions in a dog with hip dysplasia.

End feel is the sensation imparted to the examiner when the end of passive ROM is encountered. The passive motion must be performed with a relatively quick movement and the end of range is bumped briskly. Cyriax²⁴

described the quality of normal and abnormal end feels, which are defined in [Table 26-1](#).

Assessment

The therapist participates in three activities while managing a patient problem: examination, treatment, and assessment. Although each is important to the provision of a complete and quality program, assessment is the keystone. Assessment involves “open-mindedness, mental agility and mental discipline, linked with a logical and methodical process of assessing cause and effect.”² The therapist is thinking and assessing to make certain that the right task is being performed and that the task is being performed correctly. The therapist must continually assess the decision-making process during examination and treatment procedures. Assessment is implemented at three points of the patient interaction. The first assessment is implemented during the initial examination of the patient, making certain that clinical signs are examined and evaluated to reach the best diagnosis and formulate an appropriate plan of care. The second assessment occurs during the treatment session. The therapist assesses the appropriateness of the treatment techniques while the patient response is evaluated. The third assessment is an analytic process to consider the entire treatment program to determine whether the patient is progressing as a result of the treatment and to determine the prognosis for resolution of the current problem.

The Clinical Environment

The clinical setting should be safe for both the patient and the therapist so that the desired outcome of the examination and treatment session will more likely be achieved. First, a quiet area that is free of distractions for the dog facilitates relaxation and cooperation. The area should also be safe for the dog and therapist. It may be necessary to muzzle the dog initially, especially with a fearful or apprehensive dog. Second, the therapist should make certain he or she is using proper body mechanics to prevent undue stress on the spine and joints of both the dog and the therapist. Third, the therapist should approach the dog from a position where the dog can see the therapist, and he or she should touch the dog with firm, but not aggressive hands. Touch the dog at a place that is not painful and gradually move toward the involved area. Finally, the affected part should be cradled or held with a firm but gentle supporting grip to encourage relaxation and trust by the dog. If at any time the dog or therapist is not relaxed, the activity should be discontinued and further assessment and correction of the situation should be pursued.

Evaluation

The basic components of the initial examination for musculoskeletal dysfunction are observation, history of the present episode, relevant past history, active and passive ROM, functional tasks, strength assessment, and palpation. After the interview and examination are complete, all data should be evaluated to determine a rehabilitation diagnosis and a plan of care. The primary problem should be assessed to determine if it is primarily a painful (acute) or stiffness problem. After these decisions are made, only then should joint mobilizations be considered as a part of the plan of care.

Treatment with Mobilization

The effectiveness of the manual therapy treatment program is affected by the environment and the skills of the therapist. Efforts must be made to facilitate relaxation of the dog to avoid excessive pain and stress to the animal. The therapist's grip should be firm enough to support the limb, but not excessive. Good body mechanics and positioning by the therapist will help to make the movements controlled and in the desired pattern.

Grades

Joint mobilization should be planned with a specific grade of mobilization in mind. Grades are assigned to the mobilizations depending on the range through which the mobilization is applied and the point in the range where it is applied. [Figure 26-1](#) shows a modification of Maitland's graded mobilizations.²

R represents resistance. *R1* is the point in passive ROM at which the therapist senses resistance from a stretch on the noncontractile structures of a joint. *R2* is the resistance felt at the end of available passive ROM.

Grade I

Grade I mobilizations are small amplitude movements performed with three to four oscillations per second. There should be no pain during the oscillations; this is a very relaxing treatment technique for the dog. The therapist should be working in the range opposite of that which causes pain and avoid end range (*R2*). For example, if pain occurs during extension, then a grade I mobilization is performed in flexion, away from any area of the range of motion that could be painful.

Grade II

Grade II mobilizations are large-amplitude movements that are performed with three to four oscillations per second. A grade II mobilization should not cause pain during the oscillations. The mobilization should be in the range opposite that of the pain and the therapist should avoid either end range (*R2*).

Grade III

Grade III mobilizations are large-amplitude movements performed with three to four oscillations per second at the end of range of the restricted motion (bumping *R2* but then moving off of any tightness). This amplitude of motion may cause slight discomfort to the patient.

Grade IV

A grade IV mobilization is a small-amplitude movement that is performed with three to four oscillations per second between *R1* and *R2*. This amplitude of motion may cause slight discomfort for the patient. A grade IV joint mobilization is indicated when a grade III is too painful or not tolerated by the patient. Even though the grade IV may be more vigorous, it is sometimes better tolerated because the joint motion is very small and it does not completely progress to *R2*.

How to Choose the Appropriate Mobilization Grade

When ROM is decreased due to **pain**:

- If the pain is treated, then ROM should increase.
- Grade I and II mobilizations should be performed.
- Grade I and II mobilizations should be performed in the pain-free range for **30 seconds** or longer. Function should be assessed after mobilization to determine whether any change has been achieved.

When ROM is decreased due to **stiffness**:

- Grade III and IV mobilizations should be used.
- Grade III and IV mobilizations should be performed in the direction of the stiffness for 60 seconds if possible.

Function should be assessed after mobilization to determine whether any change has been achieved.

If both **pain and stiffness** are present, the therapist must decide what the primary problem is. Does the pain limit ROM or does the stiffness cause the pain? The sequence of pain and resistance can contribute to the treatment plan. For example:

- If **pain occurs before resistance**, use techniques to control the pain before progressing to more aggressive treatment.
- If **pain occurs with resistance**, mobilizations may be used with caution. However, it is customary to treat the pain and then the stiffness.
- If **pain occurs after the resistance**, vigorous mobilization may be used to treat the stiffness followed by techniques for pain.

There are two types of mobilizations:

- **Physiologic mobilizations** are done in the same pattern of movement that is produced by voluntary muscle contraction.
- **Accessory mobilizations** are performed in a pattern of movement that must be done passively and cannot be produced by voluntary muscle contraction. These should be done in the midrange of the joint motion (open pack position) because this allows for more joint play or motion.

There are additional guidelines to help the therapist decide which type of mobilization to perform.

- If pain is being treated and **physiologic mobilization** grades I and II cause pain, then **accessory** grades I and II should be performed.
- If 50% of physiologic active ROM is not available secondary to pain, then use **accessory** grades I and II to treat the pain (50/50 rule).
- If treating stiffness (Grades III and IV), the therapist should perform 60 seconds of an **accessory** mobilization followed by 60 seconds of a **physiologic** mobilization and then assess the effect. During any mobilization treatment, an assessment should be made after each set of oscillations, whether for pain or stiffness, to determine the effect of the mobilization. This sequence should be performed three times.

Oscillation versus Sustained Stretch

Mobilization for joint restrictions should be oscillations. Mobilization for muscle, tendon, and skin restrictions should be sustained. Repetitive low-intensity stretches are beneficial to stimulate tissue elongation. Contracted tissues respond positively to sustained stretch.³⁰

Open versus Closed Kinetic Chain

Dutton defines open kinetic chain activities as when the “involved end segment of an extremity [is] moving freely through space, resulting in isolated movement of a joint.”²³

Examples of an open kinetic chain activity in a dog include when a paw is swinging forward in the gait cycle or when the dog is moving a forelimb while in lateral recumbency. On the other hand, closed kinetic chain activity occurs when the distal segment of an extremity is in contact with a surface and the limb is weight bearing.²⁸ Examples of a closed kinetic chain activity include the stance phase of gait. It is of value to note whether a limb is being exercised in the open versus closed kinetic chain. Closed kinetic chain activities facilitate proprioception, muscles working in groups, and coordination of joint function. On the other hand, it is easier to achieve isolated joint motion in the open kinetic chain.

Dogs perform a large array of movements to maintain function. The restoration of normal movement plays a significant role in the treatment of the dog. Joint mobilization offers one method of obtaining the motion required for the restoration of function through manual therapy skills.

Prior to beginning mobilization on the dog, it is essential that the therapist be completely familiar with the bony landmarks associated with the joint. The joint should be evaluated to be certain that pathologic joint luxation or subluxation is not present. Hair may need to be pulled out of the way or clipped to expose the joint. Active and passive ROM should be functionally assessed and measured by goniometry prior to beginning the treatment. The assessment of passive and active ROM is an important aspect of the treatment process. The comparison of ROM before and after treatment demonstrates the effectiveness of the treatment.

The initial treatment of a dog's joint depends largely on the nature of the problem, in particular, pain or stiffness. If the primary problem is pain, a relaxation technique, such as light massage, may be used to calm the dog and to gain trust. If the animal is not in pain, but has stiffness, the joint or joints to be mobilized should be prepared for treatment with a warming modality or light exercise. Moist heat, massage, or therapeutic ultrasound may be used prior to treatment. Active exercise, including aquatic therapy, may also be used. The joint mobilization treatment should then be followed with passive and active ROM accentuating the motion addressed. Therapeutic exercises prescribed should also address the motion. Ice may then be applied after the session to help prevent soreness from the mobilization and other activities.

Forelimb injuries are common in dogs. Elbow dysplasia and resultant osteoarthritis of the elbow occurs quite frequently in dogs. In agility and jumping events, problems in the shoulder, elbow, and carpus are common secondary to the stresses placed on the forelimbs. Functional ROM at the carpus and elbow is essential to perform activities. For example, full elbow ROM is required for normal function. ROM demands in working and athletic dogs are greater than in pet dogs. For example, full shoulder extension is

required for dogs participating in agility activities such as jumping.

The hindlimb is responsible for generating propulsion of the dog. Decreased ROM may result in a loss of power and function, especially loss of joint extension. Decreased joint motion in the hindlimb is common with hip dysplasia, cruciate disease, muscle injuries, and osteoarthritis.

The Shoulder Joint

The shoulder, or glenohumeral joint, is a ball-and-socket joint and follows the convex on concave rule of motion (Box 26-1). Impairment of shoulder ROM may be seen after a fracture of the humerus, bicipital tenosynovitis, osteochondritis dissecans, infraspinatus contracture, scapular fracture, surgical repair of shoulder instability, and other conditions. Compensations in the shoulder complex may also be seen as a result of hindlimb or spinal problems such as intervertebral disk disease, canine hip dysplasia, cruciate disease, or neurologic problems. It is believed that dogs increase the amount of weight placed on their forelimbs to compensate for a problem affecting the spine or hindlimb. Over time, problems may develop in the forelimbs secondary to the increased forces placed on the joint.

Lateral Distraction of the Glenohumeral Joint (Figure 26-2)

Dog position: The dog is in lateral recumbency with the shoulder to be treated facing the therapist.

Therapist position: The therapist is ventral to the dog's shoulder, facing the shoulder to be mobilized.

Hand position: The stabilizing hand is placed on the proximal scapula. The web space and fingers of the mobilizing hand are placed under the axilla, as close to the proximal humerus as possible.

Box 26-1 Brief Overview of the Mechanics of the Shoulder Joint

- As the glenohumeral joint extends, the humeral head moves in a caudal direction. Therefore to increase extension, a caudal glide or mobilizing force should be applied.
- As the glenohumeral joint flexes, the humeral head moves in a cranial direction. Therefore to increase flexion, a cranial glide or mobilizing force should be applied.
- As the glenohumeral joint abducts, the humeral head moves in a medial and ventral direction. Therefore to increase abduction, a medial and ventral glide or mobilizing force should be applied.
- As the glenohumeral joint adducts, the humeral head moves in a lateral and dorsal direction. Therefore to increase adduction, a lateral and dorsal glide or mobilizing force should be applied.

Mobilization: A lateral distraction is applied with the mobilizing hand while the stabilizing hand maintains the position of the scapula. The distraction is held for up to 15 seconds and then released. This may be repeated up to 10 times. This is a sustained mobilization.

Objective: Stretch the capsule, relax the surrounding tissues, and prepare the joint for additional mobilizations.

This technique can be performed as a grade II mobilization to diminish pain in the shoulder joint. The mobilizing hand should be oscillated at a rate of two to three repetitions per second. The head of the humerus should be moved laterally only a minimal amount, making certain that no joint resistance is felt by the therapist and no pain is experienced by the dog. The oscillations are maintained for approximately 30 seconds, at which time the dog should be assessed to determine whether there has been any change in function. The technique may be repeated up to three times if the dog's pain improves.

In addition to a lateral distraction, a grade III oscillation can be performed to improve joint restriction. In this case, pain is not a major factor and should only be present when the joint is moved to the point of joint motion limitation. To perform a grade III mobilization, the head of the humerus is oscillated to the point of end feel of the capsule, where the bone cannot be moved any further and the therapist feels a distinct stop in the available motion. These oscillations are maintained for up to 60 seconds, assuming the dog remains relaxed and allows this motion to occur. This mobilization can be followed with a physiologic motion to improve shoulder flexion, extension, or rotation. For example, the dog's shoulder should be passively moved through physiologic, or the normal range, of flexion. This may be followed with a movement of physiologic extension and then internal and external rotation. This technique is contraindicated if medial shoulder instability is present at the shoulder. A lateral distraction may cause pain and increase instability.

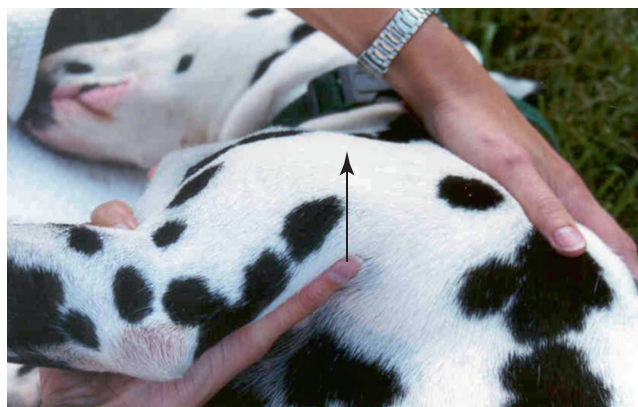


Figure 26-2 Lateral distraction of the glenohumeral joint. The arrow indicates the direction of the mobilization.

Caudal Mobilization: Accessory Glide (Figure 26-3)

Dog position: The dog is in lateral recumbency with the shoulder to be mobilized facing toward the therapist.

Therapist position: The therapist is cranioventral to the shoulder, facing the shoulder.

Hand position: The stabilizing hand is placed over the scapula; the dog's forelimb may rest on the forearm of the stabilizing hand. The thenar eminence or the web space of the mobilizing hand is placed over the greater tubercle of the humerus.

Mobilization: A gentle lateral distraction should be performed prior to the caudal glide to facilitate movement. A caudal glide is applied to move the humerus in relation to the scapula. This glide should be sustained 3 to 5 seconds.

Objective: Increase shoulder extension.

A grade III accessory mobilization may be performed from the same grips by adding a oscillating movements with the mobilizing hand. The caudal oscillations are performed with the joint held in the loose packed position, and the oscillations should bump against the end of the available joint motion. The oscillations are performed at a rate of three or four repetitions per second and are continued for up to 60 seconds. The caudal glide is typically followed with a grade III physiologic shoulder extension mobilization, oscillating for an additional 60 seconds.

Physiologic Shoulder Extension

To perform a grade III physiologic shoulder extension mobilization, the mobilizing hand and the stabilizing hand should exchange positions so that the therapist's mobilizing hand moves the humerus cranially into shoulder extension while the stabilizing hand holds the scapula stable.

Hand position: The mobilizing hand should grip the mid-shaft of the humerus.

Mobilization: With the scapula stabilized, the humerus is pushed into extension to the end of available ROM, bumping end feel of the joint. The humerus is moved back to near midposition of the joint. An oscillatory movement is performed by moving the humerus to end feel and back to midposition at a rate of three or four repetitions per second. The end feel should be felt by the therapist with each oscillation. The mobilization is performed for up to 60 seconds.

The dog's shoulder extension should be assessed for improvement in ROM. If motion is improved, the grade III accessory mobilization and the grade III physiologic mobilization can be repeated up to three times in one treatment session.

Cranial Mobilization: Accessory Glide (Figure 26-4)

Dog position: The dog is in lateral recumbency with the shoulder to be mobilized facing toward the therapist.

Therapist position: The therapist is on the ventral side of the dog, caudal to the forelimb.

Hand position: The stabilizing hand is placed over the scapula; the dog's forelimb may rest on the forearm of the stabilizing hand. The thenar eminence or web space of the mobilizing hand is placed on the caudal portion of the humerus as close to the joint as possible.

Mobilization: A cranial glide is applied to the humerus.

Objective: Increase shoulder flexion.

The cranial glide is typically followed with a grade III physiologic shoulder flexion mobilization, oscillating for an additional 60 seconds.



Figure 26-3 Caudal mobilization of the humerus to increase shoulder extension. The arrow indicates the direction of the mobilization.



Figure 26-4 Cranial mobilization of the humerus to increase shoulder flexion. The arrow indicates the direction of mobilization.

Physiologic Shoulder Flexion

To perform a grade III physiologic shoulder flexion mobilization, the mobilizing hand and the stabilizing hand should exchange positions so that the therapist's mobilizing hand moves the humerus caudally into shoulder flexion while the stabilizing hand holds the scapula stable.

Hand position: The mobilizing hand should grip the mid-shaft of the humerus.

Mobilization: With the scapula stabilized, the humerus is pushed into flexion to the end of available ROM, bumping end feel of the joint. The humerus is moved back to near midposition of the joint. An oscillatory movement is performed by moving the humerus to end feel and back to midposition at a rate of three or four repetitions per second. The end feel should be felt by the therapist with each oscillation. The mobilization is performed for up to 60 seconds.

The dog's shoulder flexion should be assessed for improvement in ROM. If motion is improved, the grade III accessory mobilization and the grade III physiologic mobilization can be repeated up to three times in one treatment session.

Other motions about the shoulder joint in the dog, including rotation, abduction, and circumduction, are important, but less crucial to the overall function of the dog. Both physiologic and accessory mobilizations to improve these motions can be performed, but are not included in this chapter.

The Elbow Joint

The elbow is a joint that is frequently problematic, with stiffness and pain secondary to degenerative joint disease. It is complex and consists of three joints—the humeroradial joint, the humeroulnar joint, and the radioulnar joint.²⁹ Clinically, stiffness in the elbow appears to respond well to distraction of the elbow joint. Pain and stiffness of the elbow may respond well to physiologic flexion and extension mobilizations. Full ROM of the elbow joint is necessary for proper function of the forelimb. Full elbow extension is necessary in the dog's normal gait cycle and a significant amount of flexion is necessary to engage in functional activities such as stair climbing, stepping over obstacles, and jumping.^{31,32}

The proximal portion of the radius is positioned laterally at the elbow, and the distal portion of the radius is positioned medially at the carpus. The articulation between the radius and ulna allows approximately 45 degrees of rotational movement of the antebrachium. The articulation between the humerus and ulna provides the hinge motion responsible for elbow flexion and extension. The proximal ulna is positioned medially at the elbow. The trochlear notch is a deep, half moon-shaped concavity that faces cranially and articulates with the convex trochlea of the



Figure 26-5 Humeroulnar distraction for the elbow. The arrow indicates the direction of mobilization.

humerus. Therefore in an open kinetic chain, the concave ulna moves on the convex humerus.

Humeroulnar Distraction (Figure 26-5)

Dog position: The dog is in lateral recumbency or in a sitting position with the elbow flexed to 90 degrees, and the radioulnar joint in slight supination. Approximately 45 degrees of flexion allows the anconeal process to move more freely in relation to the humerus without engaging the olecranon fossa.

Therapist position: The therapist is on the ventral side of the dog, facing the elbow to be mobilized.

Hand position: The stabilizing hand is located over the distal humerus. The mobilizing hand cradles the dog's forearm to maintain flexion and slight supination while grasping the proximal radius and ulna.

Mobilization: A longitudinal distraction is applied and maintained. In addition, the elbow may be moved into flexion as the distraction occurs.

Objective: Decrease joint stiffness for elbow flexion or extension.

Grade III physiologic elbow flexion or extension mobilizations may be performed following humeroulnar distraction, oscillating for an additional 60 seconds. Elbow ROM should be reassessed after the mobilization. If there is an improvement, three additional sets of the mobilization may be performed in a treatment session.

Physiologic Elbow Extension

Dog position: Lateral recumbency is preferred, although the mobilization may be performed while the dog is in a sitting position.

Therapist position: The therapist is ventral, facing the cranial aspect of the elbow to be mobilized.

Hand position: The stabilizing hand is wrapped around the distal portion of the humerus, placing the thenar and hypothenar eminences on the caudal aspect of the humerus, taking care to avoid the olecranon. The index

finger, third finger, and thumb of the mobilizing hand gently grasp the proximal radius and ulna.

Mobilization: To perform a grade III physiologic elbow extension mobilization, the therapist's mobilizing hand pushes the radius and ulna caudally into elbow extension while the stabilizing hand holds the distal humerus. With the humerus stabilized, the radius and ulna are pushed into extension to the end of available ROM, bumping the end feel of the joint. The radius and ulna are moved back to near midposition of the joint. An oscillatory movement is performed by moving the radius and ulna to end feel and back to midposition at a rate of three or four repetitions per second. End feel should be felt by the therapist with each oscillation. The mobilization is performed for up to 60 seconds.

Objective: Increase elbow extension.

Elbow ROM should be reassessed after the mobilization. If there is an improvement, three additional sets of the mobilization may be performed in a treatment session.

Physiologic Elbow Flexion

Dog position: Lateral recumbency is preferred, although the mobilization may be performed while the dog is in a sitting position.

Therapist position: The therapist is ventral to the elbow and faces the cranial aspect of the elbow to be mobilized.

Hand position: The stabilizing hand is placed on the cranial aspect of the distal humerus. The web space of the mobilizing hand is placed over the proximal radius and ulna.

Mobilization: To perform a grade III physiologic elbow flexion mobilization, the therapist's mobilizing hand pushes the radius and ulna cranially into elbow flexion while the stabilizing hand holds the distal humerus. With the humerus stabilized, the radius and ulna are pushed into flexion to the end of available ROM, bumping end feel of the joint. The radius and ulna are moved back to near midposition of the joint. An oscillatory movement is performed by moving the radius and ulna to end feel and back to midposition at a rate of three or four repetitions per second. End feel should be felt by the therapist with each oscillation. The mobilization is performed for up to 60 seconds.

Objective: To increase elbow flexion.

Elbow ROM should be reassessed after the mobilization. If there is an improvement, three additional sets of the mobilization may be performed in a treatment session.

The Carpus

Despite its intricacy, the carpus functions primarily as a hinge joint, permitting flexion and extension.²⁹ The main movement occurs at the antebrachioacarpal joint (Box 26-2), which consists of the distal radius and ulna,

Box 26-2 Overview of Kinematics of the Antebrachioacarpal Joint

- As the antebrachioacarpal joint extends in an open kinetic chain, the convex articular surface of the radial and ulnar carpal bones moves in a caudal direction on the concave articular surface of the distal radius and ulna. Therefore to increase carpal extension, a caudal glide of the radial and ulnar carpal bones on the distal radius and ulna should be applied.
- As the joint extends in a closed kinetic chain, the concave surface of the radius and ulna moves in a cranial direction on the proximal row of carpals.
- As the antebrachioacarpal joint flexes in an open kinetic chain, the convex articular surface of the radial and ulnar carpal bones moves in a cranial direction on the concave articular surface of the distal radius and ulna. Therefore to increase carpal flexion, a cranial glide should be applied.
- As the joint flexes in a closed kinetic chain, the concave radius and ulna moves in a caudal direction on the convex surface of the proximal row of carpals



Figure 26-6 A caudal accessory glide of the antebrachioacarpal joint. The arrow indicates the direction of mobilization.

and the radial and ulnar carpal bones. The distal radius and ulna offer a concave surface for the convex radial and ulnar carpal bones to move within.

Caudal Glide of the Antebrachioacarpal Joint: Accessory Glide (Figure 26-6)

Dog position: The dog is either in lateral recumbency or in a sitting or sternal position.

Therapist position: The therapist is ventral, facing the carpus to be mobilized.

Hand position: The stabilizing hand is placed over the caudal surface of the distal radius and ulna, just proximal to the joint. The mobilizing hand is placed on the dorsal portion of the radial and ulnar carpal bones.

Mobilization: A caudal mobilization is applied by directing the radial and ulnar carpal bones in a palmar direction while stabilizing the distal radius and ulna. This glide should be held for 3 to 5 seconds.

Objective: Increase carpal extension.

A grade III accessory mobilization may be performed with the same hand position by adding an oscillation with the mobilizing hand. The caudal glide mobilizations are performed with the carpus held in a loose packed position, or slight flexion, and the caudal glide oscillations should bump against the end of the available motion. The oscillations are performed at a rate of three or four repetitions per second and are continued for up to 60 seconds. The accessory glide is typically followed by a grade III physiologic carpal extension mobilization for up to 60 seconds.

Physiologic Carpal Extension Mobilization

To perform a grade III physiologic carpal extension mobilization, the mobilizing hand is placed over the palmar portion of the radial and ulnar carpal bones and the stabilizing hand is placed on the cranial surface of the distal radius. The mobilizing hand pushes the distal radial and ulnar carpal bones dorsally into carpal extension. The distal radius and ulna are stabilized while the mobilizing hand moves the distal radial and ulnar carpal bones to the end of the available ROM and then back to the mid position of the joint. This oscillation between the available end range and the mid range should be performed at a rate of three to four oscillations per second for up to 60 seconds.

As always, the ROM should be assessed for an improvement. Further mobilizations may be repeated up to three times in one treatment session. Functional activities may then be prescribed to the dog to assist in maintaining the improved ROM, such as walking and wheelbarrowing, to encourage carpal extension while in a weight-bearing position.

Cranial Glide of the Antebrachicarpal Joint (Figure 26-7)

Dog position: The dog is either in lateral recumbency or in a sitting or sternal position.

Therapist position: The therapist is ventral, facing the carpus to be mobilized.

Hand position: The stabilizing hand is placed over the cranial aspect of the distal radius and ulna. The mobilizing hand is placed on the palmar portion of the radial and ulnar carpal bones, just distal to the accessory carpal pad.

Mobilization: A cranial glide mobilization is applied by directing the radial and carpal bones dorsally while stabilizing the distal radius and ulna.

Objective: Increase carpal flexion.

A grade III accessory mobilization may be performed using the same hand positions by adding an oscillation of the mobilizing hand. The cranial glide mobilizations are



Figure 26-7 Cranial antebrachicarpal glide. The arrow indicates the direction of mobilization.

performed with the joint held in a slight amount of flexion, or the loose packed position. The oscillations should bump against the end of the available motion. The oscillations are performed at a rate of three to four repetitions per second, and are continued for up to 60 seconds. The cranial glide is then typically followed with a grade III physiologic carpal flexion mobilization for an additional 60 seconds.

Physiologic Carpal Flexion Mobilization

To perform a physiologic carpal flexion mobilization, the mobilizing hand is placed on the dorsal aspect of the radial and ulnar carpal bones and the stabilizing hand is placed on the palmar surface of the distal radius and ulna. With the distal radius and ulna stabilized, the mobilizing hand moves the radial and ulna carpal bones in a palmar direction into flexion. The radial and ulnar carpal bones are moved from the end feel of the joint back to the midposition of the joint at a rate of three or four repetitions per second for up to 60 seconds.

Carpal flexion should be assessed after the mobilization for an improvement. If the motion is improved, the grade III accessory and physiologic mobilizations may be repeated up to three times in one treatment session. Activities to be performed after mobilizations to maintain motion include high stepping over objects to encourage active flexion, or walking in water, sand, snow, or tall grass.

The Hip Joint

The hip or the coxofemoral joint is a ball-and-socket joint (Box 26-3). The convex-on-concave mobilization principles apply to this joint, similar to other joints.

Caudal Glide: Accessory Glide (Figure 26-8)

Dog position: The dog is in lateral recumbency with hip in slight abduction and slight flexion.

Box 26-3 Brief Overview of the Mechanics of the Hip Joint

- As the hip moves into flexion, the femoral head moves caudally with respect to the acetabulum. Therefore to increase hip flexion, a caudal glide is applied to the femoral head.
- As the hip moves into extension, the femoral head moves cranially with respect to the acetabulum. Therefore to increase hip extension, a cranial glide is applied to the femoral head.
- As the hip moves into abduction, the femoral head moves medially and ventrally in a rolling manner with respect to the acetabulum. Therefore to increase hip abduction, a medial and ventral glide is applied to the femoral head.
- As the hip moves into adduction, the femoral head moves laterally and dorsally in a rolling manner with respect to the acetabulum. Therefore to increase adduction, a lateral and dorsal glide is applied to the femoral head.



Figure 26-8 Caudal glide to increase hip flexion. The arrow indicates the direction of mobilization.

Therapist position: The therapist faces the hip to be mobilized on the ventral surface.

Hand position: The stabilizing hand is placed over the sacrum. The forearm of the mobilizing hand may be placed under the hindlimb to assist in guiding the limb. The web space of the mobilizing hand is placed at the cranial portion of the proximal femur, as close to the joint line as possible.

Mobilization: A caudal glide is applied with the mobilizing hand. The forearm may assist the hindlimb into flexion as the glide is applied.

Objective: Increase hip flexion.

A grade III accessory mobilization may be performed from the same described hand position by adding



Figure 26-9 Physiologic hip flexion. The arrow indicates the direction of mobilization.

an oscillation of the mobilizing hand. The caudal glide oscillations are performed with the joint held in slight flexion and slight abduction, or the loose packed position. The oscillations should bump against the end of the available motion at a rate of three or four repetitions per second and be continued for up to 60 seconds. The caudal glide is typically followed by a grade III physiologic hip flexion mobilization for an additional 60 seconds.

Physiologic Hip Flexion (Figure 26-9)

To perform a grade III physiologic hip flexion mobilization, the mobilizing hand should be moved to the caudal portion of the proximal femur, just below the ischial tuberosity. The stabilizing hand should remain on the sacrum and cradle the distal portion of the hindlimb if possible. With the pelvis stabilized, the femur is moved to the end of the available ROM and then back to near midposition of the joint. An oscillatory movement is performed by moving the femur to end feel and back to midposition at a rate of two to three repetitions per second for up to 60 seconds. With the physiologic mobilization, it is important that the therapist move the limb with his or her body instead of with arm movements only because this is a larger joint and requires proper body mechanics to avoid injury to the therapist.

Hip flexion should then be assessed after the mobilization to determine if there has been an improvement. If the motion is improved, the grade III accessory mobilization and the grade III physiologic mobilization may be repeated up to three times in one treatment session. Active movement after treatment includes high stepping, swimming, or walking in deep water to encourage active hip flexion.



Figure 26-10 Cranial mobilization to increase hip extension. The arrow indicates the direction of mobilization.

Cranial Glide: Accessory Glide (Figure 26-10)

Dog position: The dog is in lateral recumbency with hip in slight abduction and slight flexion.

Therapist position: The therapist is caudal and ventral to the dog, facing the hip to be mobilized.

Hand position: One hand may be placed at the distal femur to assist with control or placed at the sacroiliac region. Because this is a large joint to mobilize with a long lever arm, it is often easiest to place the hand not performing the mobilization at the distal femur. The pelvis is stabilized by the dog's weight in a lateral recumbent position. The forearm of the stabilizing hand may be used to assist the positioning of the hindlimb. The web space of the mobilizing hand is placed under the ischial tuberosity, on the proximal and caudal aspect of the femur.

Mobilization: A cranial mobilization is applied by directing the proximal femur cranially, with a slight dorsal angulation.

Objective: Increase hip extension.

A grade III accessory mobilization may be performed with the same hand positions as described previously. The cranial glide oscillation is performed with the hip in a loose packed position, or in slight flexion and abduction. The oscillations should bump up against the end of the available motion. The oscillations are performed at a rate of three to four repetitions per second and are continued for up to 60 seconds. The cranial glide of the femur is typically followed by a grade III physiologic hip extension mobilization for an additional 60 seconds.

Physiologic Hip Extension (Figure 26-11)

To perform the physiologic mobilization for hip extension, the mobilizing hand should be placed on the cranial and most proximal portion of the femur and the stabilizing hand should remain on the sacrum. The mobilizing hand



Figure 26-11 Physiologic hip extension. The arrow indicates the direction of mobilization.

applies a caudal force to the proximal femur to encourage hip extension. With the pelvis stabilized, the proximal femur is pushed into the available amount of extension, bumping the end feel of the joint. The femur is then moved back to the near midposition of the joint. Oscillatory movements are performed by moving the femur to the end feel and back to midposition at a rate of three to four repetitions per second for 60 seconds.

Hip extension should then be assessed after the mobilization for an improvement in ROM. If the motion is improved, the grade III accessory and physiologic mobilization should be repeated up to three times in one treatment session. Active hip extension should also be encouraged through resisted walking by having the dog pull against a leash or climbing stairs.

The Stifle Joint

The femoral condyles are a convex surface that articulate with the concave tibial plateau.²⁹ The stifle is a ginglymus joint, or a hinge joint, and flexion and extension are the primary motions at the joint (Box 26-4). The concave tibia moves on the convex femur and is therefore following the concave on convex rule. Therefore with stifle extension, the tibia glides cranially on the femur. With stifle flexion, the tibia glides caudally on the femur.

Restoration of both stifle flexion and extension is essential to normal ambulation and function in the dog. The dog requires full stifle flexion to obtain a normal sitting posture and full stifle extension for proper ambulation.

Femorotibial Cranial Glide (Figure 26-12)

Dog position: The dog is in lateral recumbency with the stifle in a resting position.

Therapist position: The therapist faces the ventral aspect of the dog and the stifle to be mobilized.

Box 26-4 Overview of Kinematics of the Stifle Joint

- As the stifle joint flexes in the open kinetic chain, the concave plateau of the tibia moves in a caudal direction on the convex femoral condyles. Therefore to increase stifle flexion, a cranial glide mobilization of the proximal tibia on the femur should be applied.
- As the stifle flexes in the closed kinetic chain, the convex femoral condyles roll caudally on the concave tibial plateaus.
- As the stifle joint extends in the open kinetic chain, the concave plateau of the tibia moves in a cranial direction on the convex femoral condyles. Therefore to increase stifle extension, a caudal glide mobilization of the proximal tibia should be applied.
- As the stifle extends in the closed kinetic chain, the convex femoral condyles roll cranially and glide caudally on the concave tibial plateaus.



Figure 26-12 Cranial glide to increase stifle flexion. The arrow indicates the direction of mobilization.

Hand position: The stabilizing hand is on the cranial surface of the distal femur. The mobilizing hand is placed on the caudal surface of the proximal tibia.

Mobilization: A cranial glide is applied to the tibia while the distal femur is stabilized.

Objective: Increase stifle joint flexion.

A grade III accessory mobilization may be performed from the same described hand position by adding an oscillation of the mobilizing hand. The cranial oscillations are performed with the joint held in slight flexion or the loose packed position. The oscillations should bump against the end of the available motion at a rate of three to four repetitions per second for up to 60 seconds. The cranial glide is typically followed by a grade III physiologic stifle flexion mobilization for an additional 60 seconds.

Because there is virtually no cranial or caudal glide motion in the normal stifle joint with intact cranial and



Figure 26-13 Physiologic stifle flexion. The arrow indicates the direction of mobilization.

caudal cruciate ligaments, the value of cranial glide and accessory joint mobilizations in the stable joint are uncertain.

Physiologic Stifle Flexion (Figure 26-13)

To perform a grade III physiologic stifle flexion mobilization, the hand positions may remain the same or the mobilizing hand may be placed on the cranial aspect of the distal tibia while the stabilizing hand is placed on the caudal aspect of the distal femur. With the femur stabilized, the tibia is moved to the end of the available range of flexion and then back to near midposition of the joint. An oscillatory movement is performed by moving the tibia to end feel and back to midposition at a rate of three or four repetitions per second for up to 60 seconds.

Stifle flexion should then be assessed after the mobilization to ascertain if there has been an improvement. If the motion is improved, the grade III accessory mobilization and the grade III physiologic mobilization may be repeated up to three times in one treatment session. Active movement after treatment includes high stepping, sit-to-stand exercises, and swimming to encourage active stifle flexion.

Femorotibial Caudal Glide (Figure 26-14)

Dog position: The dog is in lateral recumbency with the stifle in a resting position.

Therapist position: The therapist faces the caudal aspect of the stifle to be mobilized.

Hand position: The stabilizing hand is placed on the caudal aspect of the distal femur. The mobilizing hand is placed on the cranial aspect of the proximal tibia.



Figure 26-14 Caudal glide to increase stifle extension. The arrow indicates the direction of mobilization.

Mobilization: A caudal glide is performed by the mobilizing hand while the distal femur is stabilized.

Objective: Increase stifle joint extension.

A grade III accessory mobilization may be performed from the same described hand position by adding an oscillation of the mobilizing hand. The caudal glide oscillations are performed with the joint held in slight flexion or the loose packed position. The oscillations should bump against the end of the available motion at a rate of three or four repetitions per second for up to 60 seconds. The caudal glide is typically followed by a grade III physiologic stifle extension mobilization for an additional 60 seconds.

Because there is virtually no cranial or caudal glide motion in the normal stifle joint with intact cranial and caudal cruciate ligaments, the value of caudal glide and accessory joint mobilizations in the stable joint are uncertain.

Physiologic Stifle Extension (Figure 26-15)

To perform a grade III physiologic stifle extension mobilization, the hand positions may remain the same, or the mobilizing hand may be placed on the caudal aspect of the distal tibia while the stabilizing hand is placed on the cranial aspect of the distal femur. With the femur stabilized, the tibia is moved to the end of the available motion of extension and then back to near midposition of the joint. An oscillatory movement is performed by moving the tibia to end feel and back to mid-position at a rate of three to four repetitions per second for up to 60 seconds.

Stifle extension should then be assessed after the mobilization to ascertain if there has been an improvement. If the motion is improved, the grade III accessory mobilization and the grade III physiologic mobilization may be repeated up to three times in one treatment session.



Figure 26-15 Physiologic stifle extension. The arrow indicates the direction of mobilization.

The Spine

The spinal region of the dog is complex in that it allows a significant amount of movement as it absorbs and distributes the forces from the four limbs. Each area of the spine (cervical, thoracic, lumbar, and coccygeal) possesses varying amounts of motion. The cervical region permits enough movement so that the dog may be able to touch any part of its body with its nose except between the scapulas. Observation of a running dog demonstrates the amount of thoracic and lumbar flexion and extension that may be present.²⁹

Movements of the spine include extension, flexion, side flexion, and rotation. The motions of each component of the spine contribute to the total movement of the entire spine. The movement at the spinal level takes place between two vertebrae. Some regions of the spine demonstrate more mobility than other areas. For example, the thoracolumbar and the lumbosacral regions are two of the most mobile areas of the spine. These are also the areas in which there are a significant number of issues.

Mobilization techniques of the canine spine must be approached with extreme caution and knowledge of canine anatomy, medical conditions of the nervous system, the diagnosis, and the anatomy of the individual breed of dog. For example, the angulation of the thoracic spine differs according to the breed of the dog and the topline.^{29,33} A Labrador retriever has a level topline. This differs from the arched topline of a greyhound or a whippet.^{29,34} The differences in angulation of toplines alters the direction of the mobilization.

Because specialized training and clinical experience are necessary to safely perform spinal mobilizations, a limited number of basic mobilization techniques for the thoracic and lumbar regions are described. The cervical region is very complex and there are many contraindications associated with cervical mobilizations. It is essential that mobilization not be attempted without a clear comprehension of cervical anatomy, the nature of the dog's problem, and the contraindications of the region. Cervical ROM may become reduced secondary to arthritic conditions of the cervical spine, after surgery for a cervical disc, or conditions related to overuse of the forelimb muscles. It is common to see restrictions in the adjacent vertebrae after a ventral slot procedure. Changes in functional characteristics may result in decreased cervical extension while walking, lifting the head up from drinking or eating from a bowl on the floor, or flexing the head and neck as the dog ascends stairs. It is important to wait until after the surgical site has healed and is stable prior to performing any manual therapy techniques.

Spinal mobilizations are applied in the dog when there is loss of motion related to pain or dysfunction. For example, a decrease in passive thoracic dorsal to ventral movements produces decreased thoracic extension. This results in a relative lack of active thoracic extension—the dog will not be able to “dance” backward or jump up onto a piece of furniture. Of course, the spinal area must be identified as the problem and other areas of concern should be ruled out. For example, the practitioner should assess if hip disease or lumbar disease is restricting extension.

Thoracic Region

The dorsal spinous processes of the thoracic spine are very narrow and long. Some dogs object to direct pressure over the spinous process. The lower thoracic region is one of the more common areas where problems secondary to increased mobility occur. Intervertebral disc disease, compensatory problems from canine hip dysplasia, and arthritic changes are common in this area. As described earlier, dogs place more weight on the forelimbs with hip and stifle disease. From a clinical point of view, the thoracic area often exhibits restrictions as a result of these compensations. Dogs experiencing signs of hip dysplasia have decreased hip extension. Changes in the motion of the spinal column occur to compensate for the reduced extension.

There are thirteen thoracic vertebrae in the dog. The eleventh vertebra is called the *anticlinal vertebrae*.³⁵ It is a transition point along the thoracic vertebrae. The spinous processes cranial to T11 point caudally, and the spinous processes caudal to T11 point cranially. Therefore when performing mobilizations to this area, it is essential to understand the topline of the dog as well as the inclination of the particular vertebral spinous process.



Figure 26-16 Dorsal to ventral glide to thoracic spine. The arrow indicates the direction of mobilization.

Thoracic Central Dorsal to Ventral Glides (Figure 26-16)

Dog position: The dog is standing or in a sitting position with the spine in neutral, with no rotation or side flexion.

Therapist position: The therapist should stand or kneel at the dog's side, with the shoulders directly over the dog. Kneeling may be necessary if the dog is on the floor or a mat.

Hand position: The thumb of the mobilizing hand is placed over the spinous process of the vertebra to be mobilized. The remaining part of the mobilizing hand may be draped over the region. The stabilizing hand may be placed under the ventral position of the thoracic region. This hand helps to detect any changes in tension of the abdominal region in response to pain or discomfort as well as provide light counterforce stabilization. The therapist's elbows should be slightly flexed to facilitate a smooth oscillation of the vertebra.

Mobilization: Depending on the angulation of the thoracic region of the dog, a dorsal-to-ventral glide is applied at an angle from approximately 90 to 75 degrees. Oscillations for a grade II mobilization should be performed very gently so that the therapist senses only minimal resistance from joint structures and the dog does not experience any pain. Grade II mobilizations should be performed for up to 30 seconds. Oscillations for grade III mobilizations should be performed so that the therapist senses the end feel of the segment. This grade of mobilization should be performed for up to 60 seconds. After each mobilization, the dog should be assessed to determine whether there has been a functional change. Changes that may occur include an increase in functional mobility, such as with side flexion of the spine while the dog lies down or reaches for a treat near the base of the tail. Additional changes in thoracic mobility include the ability to ascend stairs with improved



Figure 26-17 Dorsal to ventral glide to lumbar spine. The arrow indicates the direction of mobilization.

extension. Pain and muscle spasms should also be diminished.

Lumbar Region

The seven lumbar vertebrae possess blunt spinous processes and the facet angles are positioned at an approximate 90-degree angle to the floor when the dog is in a standing position. Therefore mobilizations will mostly be performed at a 90-degree angle or perpendicular to the floor.

The areas of greatest mobility in the lumbar spine include the thoracolumbar region and the lumbosacral region.^{29,30} This is also an area where intervertebral disk injuries are commonly seen.³⁶

Lumbar Dorsal to Ventral Glide (Figure 26-17)

Dog position: The dog stands with the spine in a neutral position.

Therapist position: The therapist stands or kneels directly over the area to be mobilized. The therapist's shoulders should be directly over the area to be mobilized.

Hand position: The dominant thumb should be placed directly over the spinous process to be mobilized and then reinforced by the nondominant thumb. The remainder of the hand should be draped over the dog. If the therapist feels comfortable with performing the mobilization with one hand, the nondominant hand may be placed under the dog's abdomen to ascertain any signs of pain or discomfort. The shoulders should be directly over the area and the elbows and wrists slightly flexed to provide proper body mechanics.

Mobilization: A dorsal to ventral glide is applied at approximately 90 degrees or perpendicular to the skin surface. Oscillations for a grade II mobilization should be performed very gently so that the therapist senses only minimal resistance from joint structures and the dog does not experience any pain. Grade II mobilizations should be performed for up to 30 seconds. Oscillations for grade III mobilizations should be performed so that

the therapist senses the end feel for the segment. This grade of mobilization should be performed for up to 60 seconds. After each mobilization, the dog should be assessed to determine whether there has been a functional change. If an increase in ROM and function has been made, the mobilization may be repeated two more times. Functional improvements include an improved ability to ascend stairs, transition from a lying position to a sitting position, and possibly jumping. A decrease in pain and muscle spasms should be felt, as well as increased comfort during palpation. Often, dogs contract their abdominal muscles when pressure is placed on a painful intervertebral segment.

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27 Massage

Amanda Sutton and Dana Whitlock

Biologic Basis

Massage has many definitions and is employed by many practitioners of various backgrounds. The working definition used in this chapter is that massage is the manipulation of the soft tissues of the body. The use of massage has been well received in the human field and, despite a paucity of research, it has continued to be developed and explored.

The word *massage* is derived from the Arabic word *mass*, which means “to press.” Many ancient civilizations developed a system of massage, which was used for various medical conditions. Just Lucas Championniet (1843-1913) used it in the management of fractures in the United States, and it was advanced by Mary McMillan, director of physical therapy at Harvard Medical School. Dr. JB Mennell described physical treatment by movement, manipulation, and massage, which was first published in 1917. There has been variable interest in massage by medical establishments in the twentieth century, but interest in complementary medicine has grown in the United States. The American Massage Therapy Association was founded in 1943 and had grown to more than 58,000 members by 2009.

Massage is now incorporated along with the medical management of human patients for pain relief and cases of decreased mobility. It is used as a prevention measure and in the management of competitive athletes. It is also incorporated into most chiropractic, physical therapy, and osteopathic manual therapy approaches.

Biomechanics of Connective Tissue

Massage must be directed toward a specific purpose and should be aimed at promoting physical and psychological change.

The main constituent of connective tissue is collagen. Its function is to resist axial tension, and it exhibits a stress-strain behavior ([Box 27-1](#)). Microscopically, collagen fibers are arranged in bundles and have a crimped appearance. Type I collagen is in the dermis and fascia and gives support and resistance to tension.

The basal lamina of epithelial tissue contains type IV collagen, which supports the epithelial cells. Hydrogen cross-links are formed between chains and between

molecules, giving stability at fibril levels, assisting in the formation of collagen fibers. They allow the tissue to function under mechanical stress.

The orientation of the fibers depends on the stresses to which the fibers are subjected; connective tissue must be pliable, yet very strong. A stress-strain curve demonstrates the behavior of biologic materials. When a longitudinal stress is applied to collagen, the tissue responds by elongation, which occurs in the toe region of the curve ([Figure 27-1](#)). Elongation occurs as a result of straightening of the crimped fibers, and probably also as a result of some interfibrillar sliding and shear of ground substance, which flows between the collagen fibers.

Massage produces most of its effects in the toe region of the stress-strain curve. Care should always be taken to not overstretch the tissues and to avoid damage to the internal structure of connective tissue fibers. Therefore the end feel of the tissues must be understood to avoid damage to connective tissues.

Permanent deformation is termed *plastic deformation*, which occurs with microfailure and disruption of collagen cross-links. The undamaged fibers absorb a greater proportion of the load as a new length is established, which reflects the balance between elastic recoil and the resistance of the water and glycosaminoglycans to compression.

Biologic materials also have viscosity, a property of fluids of resistance to flow, and elasticity, a property of solids. Together, these result in the viscoelastic properties of tissues. The response of tissues to loading depends on how quickly the load is applied or removed. The faster the repetitive loading and unloading, the stiffer the material behaves. With rapid loading, friction results and tissue temperature rises. Energy is dissipated as heat when the tissue is returned to its original length. This is referred to as *hysteresis*.

This is relevant in the application of massage because it is important to be sensitive to the stiffness of tissues. If a low constant or repetitive load is maintained over a long period, the elongation of tissues that occurs is by creep. Conversely, if the tissue length is held constant, elongation occurs by relaxation. Massage produces low, repetitive loads over the medium term and may cause a nonpermanent creep response. Rapid loading of tissues during